SQL Joins

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Assignment Name: SQL Joins Assignment
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Creating given dummy tables in MySQL Workbench using given data :

Table 1: Customers

```
CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(50),
    City VARCHAR(50)
);

INSERT INTO Customers
VALUES
(1,'John Smith','New York'),
(2,'Mary Johnson','Chicago'),
(3,'Peter Adams','Los Angeles'),
(4,'Nancy Miller','Houston'),
(5,'Robert White','Miami');
```

Table 2: Orders

```
CREATE TABLE Orders (
    OrderID INT PRIMARY KEY,
    CustomerID INT,
    OrderDate DATE,
    Amount INT
);

INSERT INTO Orders
VALUES
(101,1,'2024-10-01',250),
(102,2,'2024-10-05',300),
```

```
(103,1,'2024-10-07',150),
(104,3,'2024-10-10',450),
(105,6,'2024-10-12',400);
```

Table 3: Payments

```
CREATE TABLE Payments (
    PaymentID VARCHAR(10) PRIMARY KEY,
    CustomerID INT,
    PaymentDate DATE,
    Amount INT
);

INSERT INTO Payments
VALUES
('P001',1,'2024-10-02',250),
('P002',2,'2024-10-06',300),
('P003',3,'2024-10-11',450),
('P004',4,'2024-10-15',200);
```

Table 4: Employees

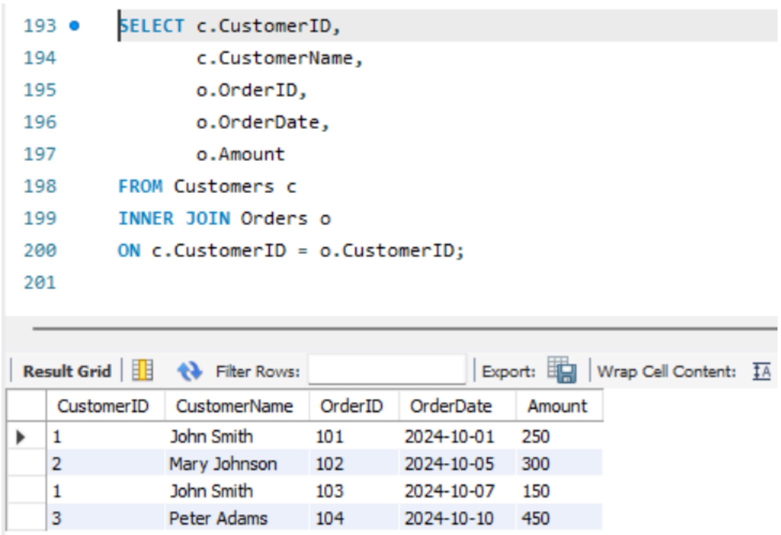
```
CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY,
    EmployeeName VARCHAR(50),
    ManagerID INT
);

INSERT INTO Employees
VALUES
(1,'Alex Green',NULL),
(2,'Brian Lee',1),
(3,'Carol Ray',1),
(4,'David Kim',2),
(5,'Eva Smith',2);
```

✔	46	00:54:50	CREATE TABLE Customers (CustomerID INT PRIMARY KEY, CustomerNa...	0 row(s) affected	0.016 sec
✔	47	00:55:03	INSERT INTO Customers VALUES (1,'John Smith','New York'), (2,'Mary Johnson'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
✔	48	00:55:09	CREATE TABLE Orders (OrderID INT PRIMARY KEY, CustomerID INT, ...	0 row(s) affected	0.094 sec
✔	49	00:55:18	INSERT INTO Orders VALUES (101,1,'2024-10-01',250), (102,2,'2024-10-05',300...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
✔	50	00:55:25	CREATE TABLE Payments (PaymentID VARCHAR(10) PRIMARY KEY, Cu...	0 row(s) affected	0.016 sec
✔	51	00:55:33	INSERT INTO Payments VALUES ('P001',1,'2024-10-02',250), ('P002',2,'2024-10...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
✔	52	00:55:39	CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, EmployeeN...	0 row(s) affected	0.016 sec
✔	53	00:55:48	INSERT INTO Employees VALUES (1,'Alex Green',NULL), (2,'Brian Lee',1), (3,'C...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.062 sec

Q1. Retrieve all customers who have placed at least one order.

Answer:

SELECT c.CustomerID,
    c.CustomerName,
    o.OrderID,
    o.OrderDate,
    o.Amount
FROM Customers c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID;



```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.OrderDate,
       o.Amount
FROM Customers c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

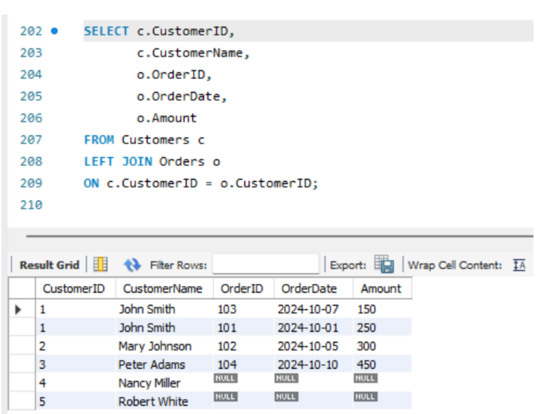
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	CustomerName	OrderID	OrderDate	Amount
1	John Smith	101	2024-10-01	250
2	Mary Johnson	102	2024-10-05	300
1	John Smith	103	2024-10-07	150
3	Peter Adams	104	2024-10-10	450

Q2. Retrieve all customers and their orders, including customers who have not placed any orders.

Answer:

SELECT c.CustomerID,
    c.CustomerName,
    o.OrderID,
    o.OrderDate,
    o.Amount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID;



```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.OrderDate,
       o.Amount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

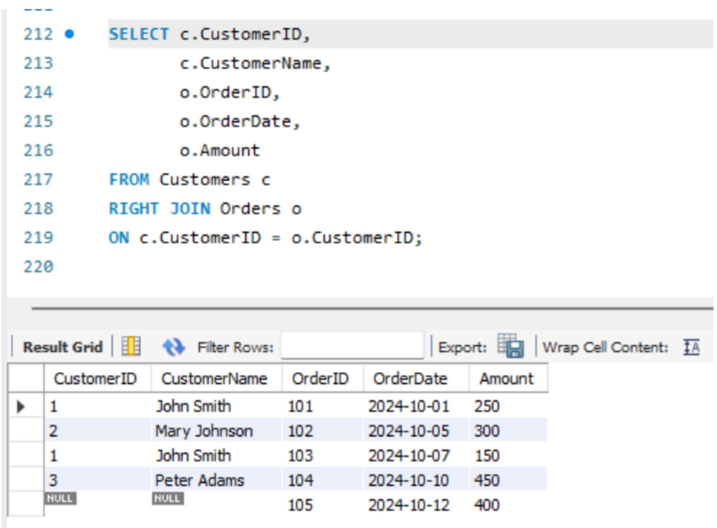
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	CustomerName	OrderID	OrderDate	Amount
1	John Smith	103	2024-10-07	150
1	John Smith	101	2024-10-01	250
2	Mary Johnson	102	2024-10-05	300
3	Peter Adams	104	2024-10-10	450
4	Nancy Miller	NULL	NULL	NULL
5	Robert White	NULL	NULL	NULL

Q3.Retrieve all orders and their corresponding customers, including orders placed by unknown customers.

Answer:

```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.OrderDate,
       o.Amount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```



```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.OrderDate,
       o.Amount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

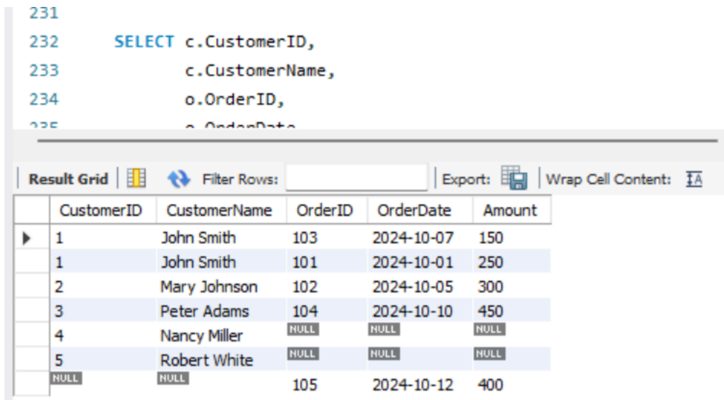
CustomerID	CustomerName	OrderID	OrderDate	Amount
1	John Smith	101	2024-10-01	250
2	Mary Johnson	102	2024-10-05	300
1	John Smith	103	2024-10-07	150
3	Peter Adams	104	2024-10-10	450
NULL	NULL	105	2024-10-12	400

Q4. Display all customers and orders, whether matched or not.

Answer:

```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.OrderDate,
       o.Amount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID

UNION

SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
```

o.OrderDate,
    o.Amount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;



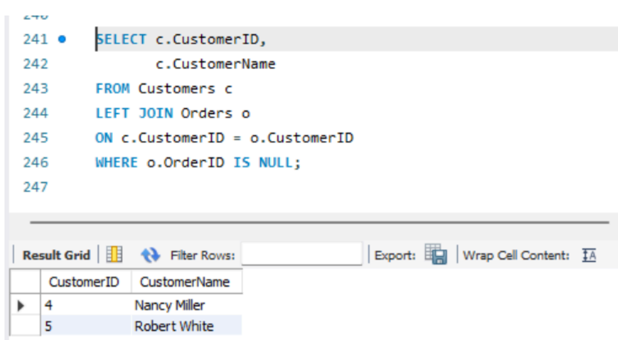
```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	CustomerName	OrderID	OrderDate	Amount
1	John Smith	103	2024-10-07	150
1	John Smith	101	2024-10-01	250
2	Mary Johnson	102	2024-10-05	300
3	Peter Adams	104	2024-10-10	450
4	Nancy Miller	NULL	NULL	NULL
5	Robert White	NULL	NULL	NULL
NULL	NULL	105	2024-10-12	400

Q5. Find customers who have not placed any orders.

Answer:
SELECT c.CustomerID,
    c.CustomerName
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;



```
SELECT c.CustomerID,
       c.CustomerName
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;
```

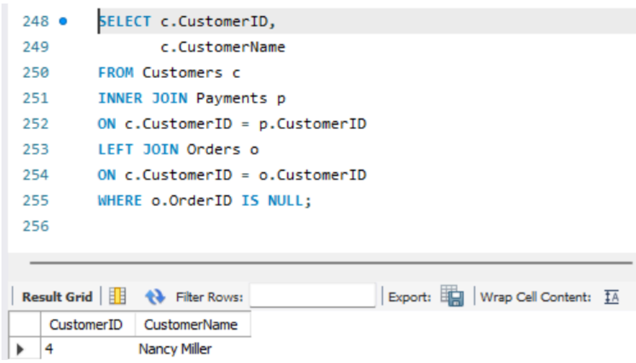
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	CustomerName
4	Nancy Miller
5	Robert White

Q6. Retrieve customers who made payments but did not place any orders.

Answer:
SELECT c.CustomerID,
    c.CustomerName
FROM Customers c
INNER JOIN Payments p

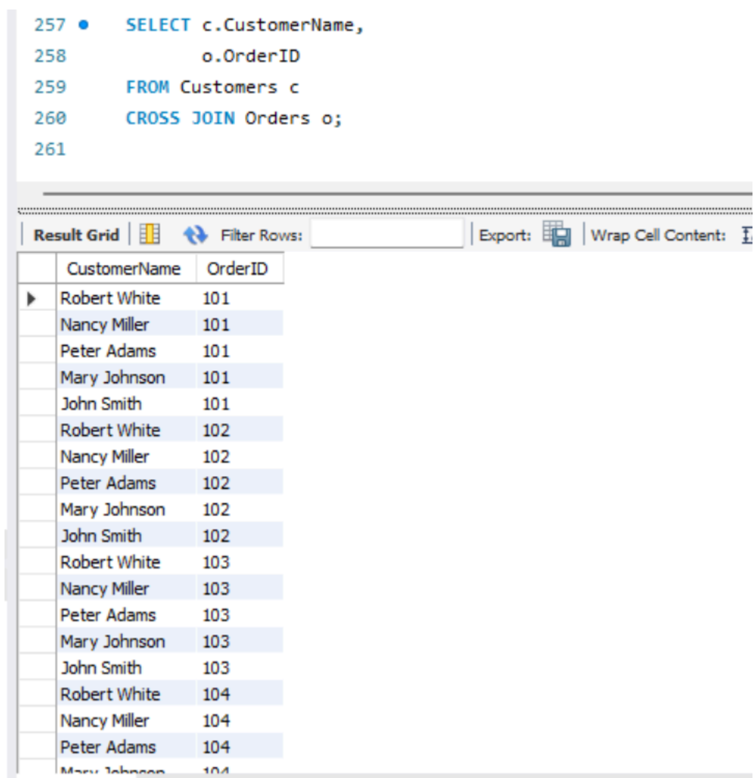
ON c.CustomerID = p.CustomerID
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;



Q7. Generate a list of all possible combinations between Customers and Orders

Answer:

```
SELECT c.CustomerName,
       o.OrderID
FROM Customers c
CROSS JOIN Orders o;
```



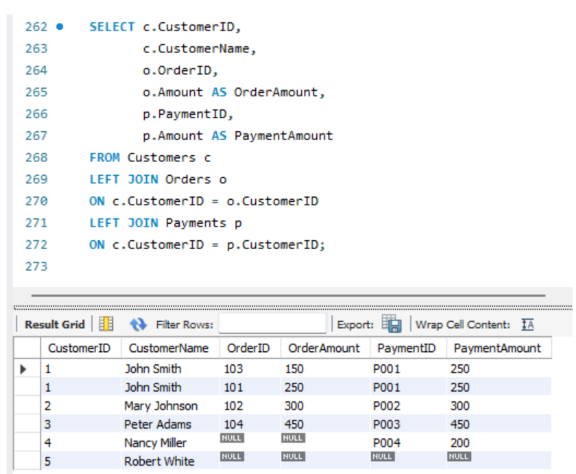
CustomerName	**OrderID**
Robert White	**101**
Nancy Miller	**101**
Peter Adams	**101**
Mary Johnson	**101**
John Smith	**101**
Robert White	**102**
Nancy Miller	**102**
Peter Adams	**102**
Mary Johnson	**102**
John Smith	**102**

Robert White	**103**
Nancy Miller	**103**
Peter Adams	**103**
Mary Johnson	**103**
John Smith	**103**
Robert White	**104**
Nancy Miller	**104**
Peter Adams	**104**
Mary Johnson	**104**
John Smith	**104**
Robert White	**105**
Nancy Miller	**105**
Peter Adams	**105**
Mary Johnson	**105**
John Smith	**105**

Q8. Show all customers along with order and payment amounts in one table.

Answer:

```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.Amount AS OrderAmount,
       p.PaymentID,
       p.Amount AS PaymentAmount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
LEFT JOIN Payments p
ON c.CustomerID = p.CustomerID;
```



```
SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.Amount AS OrderAmount,
       p.PaymentID,
       p.Amount AS PaymentAmount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
LEFT JOIN Payments p
ON c.CustomerID = p.CustomerID;
```


CustomerID	CustomerName	OrderID	OrderAmount	PaymentID	PaymentAmount
1	John Smith	103	150	P001	250
1	John Smith	101	250	P001	250
2	Mary Johnson	102	300	P002	300
3	Peter Adams	104	450	P003	450
4	Nancy Miller	NULL	NULL	P004	200
5	Robert White	NULL	NULL	NULL	NULL

Q9. Retrieve all customers who have both placed orders and made payments.

Answer:

```
SELECT DISTINCT
	c.CustomerID,
	c.CustomerName
FROM Customers c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID
INNER JOIN Payments p
ON c.CustomerID = p.CustomerID;
```